UM-2 field equations

The coupled field equations of Einstein-Lovelock gravity, the dirac16complex field, the gauge fields G, W^+, W^-, W^3 (Z, γ), B, three generations of quarks and leptons and the tip sigma field, with every interaction, and the flavour evolution of the three neutrinos

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1 How to read this document

This document derives and states the complete set of coupled field equations of the universe model UM-2: the gravitational equations with every source, the equations of every matter field with every interaction, their boundary conditions, the exact background (order-0) equations, the exact reductions to the hidden coordinate x_0 that give the particle masses, the Kohn-Sham equations of dirac16complex coupled to all the other fields, and the linear (order ϵ) equations that describe the CMB, the CNB and the BAO, and the equations of the flavour evolution of the three neutrinos in vacuum, in matter, in the early Universe and in dense neutrino gases. It is self-contained: every equation it uses is written here. Labels: AUTHOR (the author's notebooks), RECORD (the Revision record of this repository), STANDARD (textbook), ASSUMED, INVENTED (new in UM-2), DERIVED (computed and verified in UM-2; the verification is stated), ESTIMATE (an order-of-magnitude evaluation), OPEN.

Version 2 (2026-10-18) has three generations of quarks and leptons, anomaly free generation by generation, with 3×3 tip coupling matrices, from which the mixing matrices follow as $U = \kappa_e^\dagger \kappa_\nu$ and $V = \kappa_u^\dagger \kappa_d$ (section 6.2); the flavour evolution of the neutrinos is section 9. The chiral tip Z portal of version 1 is removed (section 2).

1.1 Coordinates, signature, units

The author's coordinates are $X = \{x_0, x_1, \ldots, x_7\}$: x_0 the hidden space-like direction, x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 three extra time-like directions. Frame metric $\eta_{ab} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$ ($a = 0, \ldots, 7$ aligned with $x_0, \ldots, x_7$); capital indices $M, N, \ldots$ run over $x_0, \ldots, x_7$, Greek μ, ν over the four observable coordinates $x_1, \ldots, x_4$ (signature $(+,+,+,-)$). $z = 6Hx_0$; one period of x_0 is $\pi/(6H)$. Units $\hbar = c = 1$; mass dimensions in eight dimensions: $[\Psi] = [\psi] = 7/2$, $[\kappa] = -6$ ($\kappa = 8\pi G_8$).

1.2 Conventions

- The energy-momentum tensor is $T_{MN} = -\frac{2}{\sqrt{|g|}} \frac{\delta S_{matter}}{\delta g^{MN}}$ (vielbein variation for spinors), so that the energy density is $\rho = -T^{x_4}{}_{x_4}$ and the pressures are $p_M = T^M{}_M$ (no sum) for $M \neq x_4$; for a diagonal source $T^M{}_N = \mathrm{diag}(p_0, p_3, p_3, p_3, -\rho, p_t, p_t, p_t)$.
- Gauge fields are in the geometric normalisation: the couplings stand in front of the kinetic terms, $D_M = \partial_M + \Omega_M - iG_M^\alpha T^\alpha - iW_M^a \frac{\tau^a}{2} - iYB_M$, with $[G_M] = [W_M] = [B_M] = 1$ and $[\hat{g}^2] = -4$. Formulas that show $\hat{g}$, $\hat{g}'$, g, g' explicitly refer to the canonically normalised fields $W^a/\hat{g}$, $B/\hat{g}'$, $G^\alpha/\hat{g}_s$, written with the same letters. The weak bosons are $W^\pm = (W^1 \mp iW^2)/\sqrt{2}$: W^+ **carries the electric charge** $+e$ **and** W^- **the charge** $-e$; $Z = \cos\theta_W W^3 - \sin\theta_W B$, $A = \sin\theta_W W^3 + \cos\theta_W B$.
- The gamma matrices are the author's real 16×16 matrices of $Cl(4,4)$, $\{\gamma^a, \gamma^b\} = 2\eta^{ab}$; $\bar{\Psi} = \Psi^\dagger C$ with $C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}$ (real, symmetric, $C^2 = 1$), $B = -iC\gamma^{x_4}$ (Hermitian, eigenvalues ± 1, eight each), $\Gamma_4 = i\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$, $S^{ab} = \frac{1}{4}[\gamma^a, \gamma^b]$. Since every $C\gamma^a$ is real and antisymmetric, $\bar{\Psi}\gamma^a\Psi$ is anti-Hermitian and the Hermitian vector currents are $i\bar{\Psi}\gamma^a\Psi$; $\bar{\Psi}\Psi$ is Hermitian.
- A delta function on the tip $y = -L$ in proper length is $\delta_{tip} = \delta(y + L)/f$.
- Notation: $\kappa = 8\pi G_8$ throughout (κ_f, with its subscript, is the tip mixing of a fermion). k denotes the block variable $1/f^2$ (k_*, $k_\pm$, δk, $G(k)$); as the index of α_k, $E_{(k)}$ and $\mathcal{L}_{(k)}$ it is the Lovelock order (written p in the reduction formula of section 5.6, where the block variable appears as well); in e^{ikx_1}, in h_j, in $\vec{k}$ and Δk and in the dispersion $\varepsilon = e^{-a_4}|k|$ it is the momentum of a mode along x_1, x_2, x_3; in the perturbation equations of section 8 it is the comoving Fourier wavenumber.
- $\Lambda = -10H^2$ is the eight-dimensional cosmological constant and Λ_4 the four-dimensional one; in the x_0 eigenvalue problems of section 6, $\Lambda = (m\,fL)^2$ is a dimensionless eigenvalue; $\Lambda_\star$ is the scale of the effective interactions; Λ_ℓ and Λ_q are the tip coupling matrices.
- n is a number density (n_o of a Kohn-Sham orbital, n_ψ of a Standard-Model fermion), except as a level number ($n^2\pi^2$, the $n \geq 1$ modes) and in the lattice vector $\vec{n} \in \mathbb{Z}^3$ of a momentum shell. The subscript t (x_t, p_t, v_t)

refers to the extra times x_5, x_6, x_7; $\xi = y/L$ is the scaled hidden coordinate; primes denote $d/d\xi$ in section 6 and $d/d\tau$ (conformal time) in section 8 ($\hat{g}'$ and c_Z' are names of couplings). f is the block scale, f_o an occupation number, and the label f in S_f and $T^M{}_N[f]$ stands for the Standard-Model fermions.

- w_k are the Lovelock weights; $w = 1/2$ is the weight of a fold parity in the Kohn-Sham sums, and $w = \sqrt{c^2 - e^2}$ in the fermion mass condition. ψ is a Standard-Model fermion field, except in section 8, where ψ and ϕ are the Newtonian-gauge potentials, and in section 9, where ψ_α are flavour amplitudes. ε is the energy of a mode or orbital.
- Flavour: $i = 1, 2, 3$ is the generation index of the fermion fields and labels the neutrino mass states (i, j are 3-space indices in the background curvatures); $\alpha \in \{e, \mu, \tau\}$ is a flavour index (a colour index on G^α, T^α, $f^{\alpha\beta\gamma}$); U is the PMNS and V the CKM matrix (the gauge transformations are U_L, U_Y, the block basis V_β, the transverse volume $V_\perp$, and the matter potentials carry subscripts, V_{CC}, V_{NC}, $V_{\nu\nu}$); κ_ν, κ_e, κ_u, κ_d are the 3×3 tip matrices; $\rho_{\vec{p}}$ (with a momentum subscript) is a neutrino density matrix; $\mu = \sqrt{2} G_F n_{\nu_e}$ (without subscript) is the unit of the neutrino self-interaction, μ_ψ a bulk mass and μ also the muon; e is the elementary charge, the electron and the fermion eigenvalue $\varepsilon f L$; s is also $\sinh w/w$, and $s_{ij} = \sin\theta_{ij}$, $c_{ij} = \cos\theta_{ij}$ (while c with the subscripts L, Q, R, ψ is a localisation $f\mu L$); J is the Jarlskog invariant (and the block operator $\gamma^{x_0}\gamma^{x_1}\gamma^{x_4}$); in section 9 $u = \cos\vartheta$ is the direction cosine of a neutrino (u is also the up quark and $u(y)$ a boson profile).
- H is the author's constant, except in the neutrino-decoupling equations of section 9, where H is the Hubble rate; the time is always x_4 (dots and d/dx_4), and the symmetry axis of the fast-flavour analysis is x_3. Λ^0, Λ^3 (upper indices) are the fast-flavour potentials. $G(k)$ is the Einstein-Hilbert coefficient and $G(u)$ the electron-lepton-number distribution; $F_h = 1.0095$ is the heating factor of ΔN_{eff} and $F(T_1, T_2)$ the energy-transfer kernel.

2 The action

$$S = S_{EL} + S_{stab} + S_\Psi + S_{gauge} + S_f + S_{tip} + S_{I11} + S_{I12}.$$

Gravity (AUTHOR weights):

$$S_{EL} = \frac{1}{2\kappa}\int d^8x\sqrt{|g|}\Big(\sum_{k=1}^{3}\alpha_k\mathcal{L}_{(k)} - 2\Lambda\Big), \qquad \alpha_k = w_k H^{2-2k}, \quad w = \Big(1, \frac{1}{32}, \frac{1}{1152}\Big), \quad \Lambda = -10H^2,$$

$$\mathcal{L}_{(k)} = \frac{1}{2^k}\,\delta^{A_1B_1\cdots A_kB_k}_{C_1D_1\cdots C_kD_k}\,R^{C_1D_1}{}_{A_1B_1}\cdots R^{C_kD_k}{}_{A_kB_k}: \qquad \mathcal{L}_{(1)} = R, \quad \mathcal{L}_{(2)} = R^2 - 4R_{AB}R^{AB} + R_{ABCD}R^{ABCD}.$$

Block stabiliser (ASSUMED; origin OPEN):

$$S_{stab} = -\int d^8x\sqrt{|g|}\,\rho_b(f), \qquad \kappa\rho_b(f) = \kappa\rho_b^*\Big(\frac{f_*}{f}\Big)^8, \qquad f = e^{-a_4}b_4^{1/2},$$

whose energy-momentum tensor is $T^M{}_N[\text{stab}] = \text{diag}(\rho_b, -\rho_b, -\rho_b, -\rho_b, -\rho_b, \rho_b, \rho_b, \rho_b)$.

dirac16complex (RECORD Lagrangian, canonical spin connection):

$$S_\Psi = \int d^8x\sqrt{|g|}\Big[\tfrac{1}{2}\big(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi\big) - m\bar\Psi\Psi - \frac{\lambda}{2}(\bar\Psi\Psi)^2\Big], \qquad m = 3HM = H,$$

$$D_M\Psi = \partial_M\Psi + \Omega_M\Psi, \quad D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M, \quad \Omega_M = \tfrac{1}{2}\omega_{Mab}S^{ab},$$
$$\omega_M{}^a{}_b = e^a{}_N\big(\partial_M e_b{}^N + \Gamma^N{}_{ML}e_b{}^L\big), \qquad \gamma^M = e_a{}^M\gamma^a.$$

Gauge fields (ASSUMED structure; $SU(3)_c \times SU(2)_L \times U(1)_Y$ in eight dimensions):

$$S_{gauge} = -\frac{1}{4}\int d^8x\sqrt{|g|}\Big(\frac{1}{\hat{g}_s^2}G^\alpha_{MN}G^{\alpha MN} + \frac{1}{\hat{g}^2}W^a_{MN}W^{aMN} + \frac{1}{\hat{g}'^2}B_{MN}B^{MN}\Big),$$

$$G^\alpha_{MN} = \partial_M G^\alpha_N - \partial_N G^\alpha_M + f^{\alpha\beta\gamma}G^\beta_M G^\gamma_N, \quad W^a_{MN} = \partial_M W^a_N - \partial_N W^a_M + \epsilon^{abc}W^b_M W^c_N, \quad B_{MN} = \partial_M B_N - \partial_N B_M.$$

Fermions (three generations; ASSUMED structure, boundary conditions INVENTED):

$$S_f = \sum_{i=1}^{3}\sum_{\psi}\int d^8x\sqrt{|g|}\Big[\tfrac{1}{2}\big(\bar\psi_i\gamma^M D_M\psi_i - (D_M\bar\psi_i)\gamma^M\psi_i\big) - \mu_{\psi i}\bar\psi_i\psi_i\Big], \qquad \psi\in\{Q,u_R,d_R,L,e_R,\nu_R\},$$

with, in every generation i, Q_i $(\mathbf{3},\mathbf{2},1/6) = (u,d)_L,(c,s)_L,(t,b)_L$, u_{Ri} $(\mathbf{3},\mathbf{1},2/3) = u_R,c_R,t_R$, d_{Ri} $(\mathbf{3},\mathbf{1},-1/3)$ $= d_R,s_R,b_R$, L_i $(\mathbf{1},\mathbf{2},-1/2) = (\nu_e,e)_L,(\nu_\mu,\mu)_L,(\nu_\tau,\tau)_L$, e_{Ri} $(\mathbf{1},\mathbf{1},-1) = e_R,\mu_R,\tau_R$ and ν_{Ri} $(\mathbf{1},\mathbf{1},0)$; $Q = T_3 + Y$. The bulk masses are diagonal in the generation index, and the doublets of a sector share one bulk mass (ASSUMED): $\mu_{Li} = \mu_L$, $\mu_{Qi} = \mu_Q$.

The tip sigma field (INVENTED; no Higgs particle), on the seven-dimensional tip $y = -L$:

$$S_{tip} = \int_{y=-L} d^7x\sqrt{|g_{tip}|}\Big[-\frac{v_\Sigma^2}{4}\mathrm{tr}\big((D_\mu\Sigma)^\dagger D^\mu\Sigma\big) - \sum_{i,j}\big(\bar L_i\,\Sigma\,(\Lambda_\ell)_{ij}\,\ell_{Rj} + \bar Q_i\,\Sigma\,(\Lambda_q)_{ij}\,q_{Rj} + \text{h.c.}\big)\Big], \qquad \Sigma\in SU(2),$$

$$D_\mu\Sigma = \partial_\mu\Sigma - iW_\mu^a\frac{\tau^a}{2}\Sigma + iB_\mu\Sigma\frac{\tau^3}{2}, \qquad \ell_{Rj} = (\nu_{Rj},e_{Rj})^T, \quad q_{Rj} = (u_{Rj},d_{Rj})^T,$$
$$(\Lambda_\ell)_{ij} = \mathrm{diag}\big((\lambda_\nu)_{ij},(\lambda_e)_{ij}\big), \qquad (\Lambda_q)_{ij} = \mathrm{diag}\big((\lambda_u)_{ij},(\lambda_d)_{ij}\big),$$

with complex 3×3 tip coupling matrices λ_ν, λ_e, λ_u, λ_d; invariant under $\Sigma\to U_L\Sigma U_Y^\dagger$, $U_Y = e^{i\theta\tau^3/2}$, for every choice of the matrices (DERIVED with random group elements and random complex matrices, deviation 1.1×10^{-14}).

Portals (INVENTED):

$$S_{I11} = \int d^8x\sqrt{|g|}\,\frac{c_V}{\Lambda_\star^6}\,(i\bar\Psi\gamma^M\Psi)\,\mathcal{J}_M, \qquad \mathcal{J}_M = \sum_{i=1}^{3}\sum_{\psi} q_\psi\,(i\bar\psi_i\gamma_M\psi_i),$$

$$S_{I12} = -\int d^8x\sqrt{|g|}\,\frac{\bar\Psi\Psi}{\Lambda_\star^3}\,\mathcal{F}, \qquad \mathcal{F} = c_G G^\alpha_{MN}G^{\alpha MN} + c_W W^a_{MN}W^{aMN} + c_B B_{MN}B^{MN}.$$

In the unitary gauge $\Sigma = 1$ the sigma-field kinetic term becomes $-\frac{v_\Sigma^2}{8}\big[2\hat g^2 W_\mu^+W^{-\mu} + (\hat g^2+\hat g'^2)Z_\mu Z^\mu\big]$ (canonical fields).

Removed in version 2. Version 1 also contained the chiral tip Z portal $S_{I9} = \int_{y=-L} d^7x\sqrt{|g_{tip}|}\,(c_Z + c'_Z\varsigma)(i\bar\Psi\gamma^\mu P_+\Psi)\,\frac{iv_\Sigma^2}{2}\mathrm{tr}(\Sigma^\dagger D_\mu\Sigma\tau^3)$ ($P_+ = (1+\Gamma_4)/2$, $\varsigma = \pm1$ the side of the tip), which let a band of dirac16complex stand in for the neutrinos missing from one generation. With three generations its even part c_Z is bounded by the Z invisible width ($N_\nu = 3 + 2r^2 = 2.9963 \pm 0.0074$ gives $r^2 < 0.0056$; DERIVED) and would still keep two neutrino-like species in equilibrium down to 14.7 MeV, adding

$$\Delta N_{eff} = n_W\Big(\frac{10.75}{g_s}\Big)^{4/3}F_h, \qquad F_h = \Big[\frac{T_\nu}{T_\gamma}\Big(\frac{11}{4}\Big)^{1/3}\Big/(1+h_\nu)\Big]^4 = 1.0095$$

($n_W = 2$ Weyl species decoupling when the plasma has g_s entropy degrees of freedom; F_h corrects for the heating of the neutrinos by e^+e^- annihilation, which an earlier-decoupled species does not receive, with $T_\nu/T_\gamma = 0.71664$ and the comoving heating $h_\nu = 0.00165$ of the common-temperature decoupling of section 9.4), i.e. $\Delta N_{eff} = 2.02$ at $g_s = 10.75$; its oriented part c'_Z would thermalise the Ψ_A fold band through $\Psi_A^{odd} \leftrightarrow \Psi_A^{fold} + Z$ and add $\Delta N_{eff} = 0.096$ to 0.126 ($g_s = 105.75$ to 86.25; ESTIMATE), 2.1 to 2.4 standard deviations above the ACT DR6 value $N_{eff} = 2.86 \pm 0.13$ (disfavoured): UM-2 sets $c_Z = c'_Z = 0$ (ASSUMED), so neither fold band of Ψ is ever thermalised and the fold parity is exact in both halves.

3 The gravitational field equations

Varying S with respect to the metric gives

$$\sum_{k=1}^{3}\alpha_k\,E_{(k)}{}^M{}_N + \Lambda\,\delta^M_N = \kappa\,T^M{}_N, \qquad E_{(k)}{}^M{}_N = -\frac{1}{2^{k+1}}\,\delta^{MA_1B_1\cdots A_kB_k}_{NC_1D_1\cdots C_kD_k}\,R^{C_1D_1}{}_{A_1B_1}\cdots R^{C_kD_k}{}_{A_kB_k},$$

$E_{(1)}$ being the Einstein tensor. Every $E_{(k)}$ is divergence-free identically (DERIVED for the order-0 metric for $k=1,2,3$), so the field equations imply $\nabla_M T^M{}_N = 0$, which holds on the matter field equations. The total source is

$$T^M{}_N = T^M{}_N[\text{stab}] + T^M{}_N[\Psi] + T^M{}_N[\text{gauge}] + T^M{}_N[f] + T^M{}_N[\text{tip}] + T^M{}_N[I11] + T^M{}_N[I12].$$

3.1 The energy-momentum tensors

Stabiliser. ρ_b depends on the metric through the proper block volume $V_b \propto f^4$, $\rho_b \propto V_b^{-2}$; the variation gives $T_{\mu\nu} = -g_{\mu\nu}\rho_b$ for $\mu,\nu \in \{x_1,\dots,x_4\}$ and $T_{AB} = +g_{AB}\rho_b$ for $A,B \in \{x_0,x_5,x_6,x_7\}$, i.e. $\rho = \rho_b$, $p_3 = -\rho_b$, $p_0 = p_t = \rho_b$.

dirac16complex (and every Standard-Model fermion with $m \to \mu_\psi$, $\lambda \to 0$, D_M with gauge fields):

$$T_{MN}[\Psi] = -\frac{1}{2}\Big[\bar\Psi\gamma_{(M}D_{N)}\Psi - \big(D_{(M}\bar\Psi\big)\gamma_{N)}\Psi\Big] + g_{MN}\,\hat{\mathcal{L}}_\Psi, \qquad \hat{\mathcal{L}}_\Psi = \tfrac{1}{2}\big(\bar\Psi\gamma^P D_P\Psi - (D_P\bar\Psi)\gamma^P\Psi\big) - mS - \frac{\lambda}{2}S^2,$$

with $X_{(MN)} = \frac{1}{2}(X_{MN}+X_{NM})$ and $S = \bar\Psi\Psi$. On shell for a homogeneous state this gives $\rho = mS+U$, $p = SU'-U$ with $U = \frac{\lambda}{2}S^2$ (the record's convention, $\rho = -T^{x_4}{}_{x_4}$).

Gauge fields:

$$\begin{aligned} T_{MN}[\text{gauge}] =& \frac{1}{\hat g_s^2}\Big(G^\alpha_{MP}G^\alpha{}_N{}^P - \frac{1}{4}g_{MN}G^\alpha_{PQ}G^{\alpha PQ}\Big) + \frac{1}{\hat g^2}\Big(W^a_{MP}W^a{}_N{}^P - \frac{1}{4}g_{MN}W^a_{PQ}W^{aPQ}\Big) \\ &+ \frac{1}{\hat g'^2}\Big(B_{MP}B_N{}^P - \frac{1}{4}g_{MN}B_{PQ}B^{PQ}\Big). \end{aligned}$$

Tip (for μ,ν tangent to the tip; zero for the y components):

$$T_{\mu\nu}[\text{tip}] = \delta_{tip}\Big[\frac{v_\Sigma^2}{2}\,\mathrm{tr}\big((D_{(\mu}\Sigma)^\dagger D_{\nu)}\Sigma\big) + g_{\mu\nu}\,\hat{\mathcal{L}}_{tip}\Big], \qquad \hat{\mathcal{L}}_{tip} = -\frac{v_\Sigma^2}{4}\mathrm{tr}\big((D\Sigma)^\dagger D\Sigma\big) - \big(\bar L\Sigma\Lambda_\ell\ell_R + \bar Q\Sigma\Lambda_q q_R + \text{h.c.}\big).$$

Portals. The frame currents $i\bar\Psi\gamma^a\Psi$, $i\bar\psi\gamma^a\psi$ and the scalars $\bar\Psi\Psi$ do not depend on the vielbein, so only $\sqrt{|g|}$ and the explicit metric factors vary:

$$T_{MN}[I11] = g_{MN}\,\frac{c_V}{\Lambda_\star^6}(i\bar\Psi\gamma^P\Psi)\mathcal{J}_P,$$

$$T_{MN}[I12] = \frac{4\bar\Psi\Psi}{\Lambda_\star^3}\Big[c_G\Big(G^\alpha_{MP}G^\alpha{}_N{}^P - \frac{1}{4}g_{MN}G^\alpha_{PQ}G^{\alpha PQ}\Big) + (W,c_W) + (B,c_B)\Big],$$

so I12 acts as a field-dependent gauge coupling, $1/\hat g_i^2 \to 1/\hat g_i^2 + 4c_i\bar\Psi\Psi/\Lambda_\star^3$.

4 The matter field equations

4.1 dirac16complex

$$\gamma^M D_M\Psi = \Big[m + \lambda\bar\Psi\Psi + \frac{\mathcal{F}}{\Lambda_\star^3}\Big]\Psi - i\,\frac{c_V}{\Lambda_\star^6}\,\mathcal{J}_M\gamma^M\Psi,$$

and its conjugate. The gravitational term is $\gamma^M\Omega_M$; for the order-0 metric (DERIVED, exact for every a_4, b_4 and on both halves of a period)

$$\gamma^M\Omega_M = \frac{3}{2}\,\frac{H}{f}\,\gamma^{x_0} + \Big(\frac{\dot b_4}{b_4} - \frac{\dot a_4}{2}\Big)\gamma^{x_4},$$

which vanishes only in flat 4+4 space ($H=0$, constant a_4, b_4): the coupling of Ψ to gravity is non-trivial. Ψ is a singlet of the Standard-Model group: it couples to the Standard Model only through I11 and I12.

4.2 The Standard-Model fermions

For a multiplet ψ_i of generation i and charge q_ψ (I11),

$$\gamma^M D_M \psi_i = \mu_{\psi i}\,\psi_i - i\,\frac{c_V}{\Lambda_\star^6}\,q_\psi\,(i\bar\Psi\gamma_M\Psi)\,\gamma^M\psi_i + \delta_{tip}\,(\text{Yukawa})_i,$$

$$(\text{Yukawa})_{L_i} = \sum_j \Sigma(\Lambda_\ell)_{ij}\,\ell_{Rj}, \qquad (\text{Yukawa})_{\ell_{Rj}} = \sum_i (\Lambda_\ell)^*_{ij}\Sigma^\dagger L_i,$$

$$(\text{Yukawa})_{Q_i} = \sum_j \Sigma(\Lambda_q)_{ij}\,q_{Rj}, \qquad (\text{Yukawa})_{q_{Rj}} = \sum_i (\Lambda_q)^*_{ij}\Sigma^\dagger Q_i.$$

For one generation the tip terms are equivalent to the self-adjoint mixing conditions at $y=-L$ (DERIVED; block form $\chi=(a,ib)$)

$$b_L(-L) = \kappa_f\,b_R(-L), \qquad a_R(-L) = -\kappa_f\,a_L(-L),$$

with κ_f the dimensionless tip mixing. For three generations they hold in each sector (neutrinos, charged leptons, up-type quarks, down-type quarks) for the vectors of boundary values over the generations,

$$b_L(-L) = \kappa\,b_R(-L), \qquad a_R(-L) = -\kappa^\dagger a_L(-L),$$

with a 3×3 tip matrix κ (κ_ν, κ_e, κ_u, κ_d), and are self-adjoint for every complex κ: the boundary form $\sum[\phi^\dagger\sigma_1\chi - \chi^\dagger\sigma_1\phi]$ of the coupled (L,R) problem over the three generations vanishes identically (DERIVED, sympy, 42 real variables). UM-2 takes (ASSUMED) $\kappa_e=\kappa_u=1$, $\kappa_\nu = U_{PMNS}$, $\kappa_d = V_{CKM}$.

4.3 The gauge fields

With the fermion currents $j_s^{\alpha N} = \sum_q i\bar q\gamma^N T^\alpha q$, $j^{aN} = \sum_{\text{doublets}} i\bar\psi\gamma^N\frac{\tau^a}{2}\psi$, $j_Y^N = \sum_\psi Y_\psi\, i\bar\psi\gamma^N\psi$, the effective inverse couplings $\frac{1}{\hat g^2_{i,eff}} = \frac{1}{\hat g_i^2} + \frac{4c_i\bar\Psi\Psi}{\Lambda_\star^3}$ (I12), and in the unitary gauge $\Sigma=1$ (geometric fields),

$$D_M\Big(\frac{1}{\hat g^2_{s,eff}}G^{\alpha MN}\Big) = j_s^{\alpha N},$$

$$D_M\Big(\frac{1}{\hat g^2_{eff}}W^{1,2\,MN}\Big) = j^{1,2\,N} + \delta_{tip}\frac{v_\Sigma^2}{4}W^{1,2\,N},$$

$$D_M\Big(\frac{1}{\hat g^2_{eff}}W^{3\,MN}\Big) = j^{3N} + \delta_{tip}\frac{v_\Sigma^2}{4}(W^3-B)^N,$$

$$\nabla_M\Big(\frac{1}{\hat g'^2_{eff}}B^{MN}\Big) = j_Y^N - \delta_{tip}\frac{v_\Sigma^2}{4}(W^3-B)^N,$$

where N runs over the tip directions in the δ_{tip} terms and the fermion currents are summed over the three generations. The tip terms are mass terms for $W^\pm$ and Z only: the combination $A=\sin\theta_W W^3+\cos\theta_W B$ (canonical fields) receives none, and the photon stays massless. For an x_0 profile these tip terms become the Robin conditions of section 6.1.

4.4 The tip sigma field

Σ has no radial mode. Its equations of motion are the Noether identities of the gauge symmetry: they follow from the divergence of the tip parts of the gauge-field equations (STANDARD for a gauged non-linear sigma model), so in the unitary gauge $\Sigma=1$ they are not independent; its three modes are the longitudinal W^+, W^- and Z.

4.5 Boundary conditions

The variational principle with the fold reflection $R: z\to\pi-z$ and the regular tip at $y=-L$ gives:

- **gravity and gauge fields**: even (Neumann) or odd (Dirichlet) at the fold, $\partial_y u=0$ or $u=0$; at the tip the natural (Robin) condition from the tip terms;

- **dirac16complex**: fold $(1-\gamma^{x_0})\Psi=0$ (even) or $(1+\gamma^{x_0})\Psi=0$ (odd), both parities being modes of the full period; tip $(1-\gamma^{x_0})\Psi=0$;
- **Standard-Model fermions**, with $P_{sel}=\frac{1+\Gamma_4\gamma^{x_0}}{2}\,\frac{1-i\gamma^{x_5}\gamma^{x_6}}{2}$: doublets ($\mu_\psi>0$) $P_{sel}(1-\gamma^{x_0})\psi=0$ and $(1-P_{sel})(1+\gamma^{x_0})\psi=0$ at the fold, $(1-\gamma^{x_0})\psi=0$ at the tip; singlets ($\mu_\psi<0$) the same with $\gamma^{x_0}\to-\gamma^{x_0}$; at the tip modified by the mixing conditions above.

Each condition is Lagrangian for the boundary form of the Dirac operator, so every operator is self-adjoint (DERIVED).

5 The background (order 0)

5.1 The metric, the variables and the curvature

The order-0 metric is the corrected primordial metric of the Revision record (version 2) in the author's coordinates,

$$\begin{aligned}\bar g=\operatorname{diag}\big(&e^{-2a_4}b_4\cot^2 z,\ e^{2a_4},\ e^{2a_4},\ e^{2a_4},\ -1,\\ &-e^{-2a_4}b_4(\sin^2 z)^{1/6},\ -e^{-2a_4}b_4(\sin^2 z)^{1/6},\ -e^{-2a_4}b_4(\sin^2 z)^{1/6}\big),\qquad z=6Hx_0,\end{aligned}$$

with $a_4(x_4)$, $b_4(x_4)>0$. With $y=\ln|\sin z|/(6H)$ ($dy=\cot z\,dx_0$) it reads

$$ds^2=e^{2a_4}d\vec{x}^{\,2}-dx_4^2+f^2\big[dy^2-e^{2Hy}d\vec{x}_t^{\,2}\big],\qquad f=e^{-a_4}b_4^{1/2},\quad k=\frac{1}{f^2},\quad A=\dot a_4,\quad h=\frac{\dot f}{f}=\frac{\dot b_4}{2b_4}-\dot a_4,$$

dots denoting d/dx_4; $\dot k=-2kh$. $\sqrt{|\bar g|}=e^{3a_4}f^4|\cot z|(\sin^2 z)^{1/4}=e^{3a_4}f^4e^{3Hy}|dy/dx_0|$. The Riemann tensor is pair-diagonal ($R^{ab}{}_{cd}=0$ unless $\{a,b\}=\{c,d\}$) and every sectional curvature $K_{ab}=R^{ab}{}_{ab}$ is one of (DERIVED, identical in both charts, independent of x_0)

$$K_{ij}=A^2,\quad K_{i4}=A^2+\dot A,\quad K_{iB}=Ah,\quad K_{4B}=h^2+\dot h,\quad K_{BB'}=h^2-H^2k,$$

$i,j\in\{1,2,3\}$, $B,B'\in\{0,5,6,7\}$. For a pair-diagonal Riemann tensor the Lovelock densities and tensors are (DERIVED; checked against the Ricci route, the explicit Gauss-Bonnet tensor, the Bianchi identities and the minisuperspace variation with a lapse)

$$\mathcal{L}_{(k)}=2^k k!\sum_{k\text{ disjoint pairs}}\prod_{\text{pairs}}K_{ab},\qquad E_{(k)}{}^{h}{}_{h}=-2^{k-1}k!\sum_{k\text{ disjoint pairs avoiding }h}\prod_{\text{pairs}}K_{ab}.$$

5.2 The Lovelock tensors of the background

Every off-diagonal component vanishes; the x_0 and the extra-time components coincide, $E_{(k)}{}^{x_0}{}_{x_0}=E_{(k)}{}^{x_5}{}_{x_5}$, and nothing depends on x_0 (DERIVED). The independent components are

$$E_{(1)}{}^{x_4}{}_{x_4}=6kH^2-12Ah-6h^2-3A^2$$

$$E_{(1)}{}^{x_1}{}_{x_1}=6kH^2-8Ah-4\dot h-10h^2-2\dot A-3A^2$$

$$E_{(1)}{}^{x_0}{}_{x_0}=3kH^2-9Ah-3\dot h-6h^2-3\dot A-6A^2$$

$$\begin{aligned}E_{(2)}{}^{x_4}{}_{x_4}=&144Ahk H^2+24kH^2h^2+72kA^2H^2-12H^4k^2-48hA^3-216A^2h^2\\&-144Ah^3-12h^4\end{aligned}$$

$$\begin{aligned}E_{(2)}{}^{x_1}{}_{x_1}=&96AhkH^2+48\dot hkH^2+72kH^2h^2+48\dot AkH^2+72kA^2H^2-96Ah\dot h\\&-32A\dot Ah-48\dot hh^2-12H^4k^2-48\dot Ah^2-32hA^3-16\dot hA^2\\&-136A^2h^2-192Ah^3-60h^4\end{aligned}$$

$$\begin{aligned}E_{(2)}{}^{x_0}{}_{x_0}=&36AhkH^2+12\dot hkH^2+12kH^2h^2+36\dot AkH^2+72kA^2H^2-72Ah\dot h\\&-72A\dot Ah-12\dot hh^2-36\dot Ah^2-108hA^3-36\dot hA^2-180A^2h^2\\&-12\dot AA^2-108Ah^3-12h^4-12A^4\end{aligned}$$

$$E_{(3)}{}^{x_4}{}_{x_4} = 864hkA^3H^2 + 1296kA^2H^2h^2 - 216A^2H^4k^2 - 1440A^3h^3 - 1080A^2h^4$$

$$\begin{aligned}E_{(3)}{}^{x_1}{}_{x_1} =& 576Ah\dot{h}kH^2 + 576A\dot{A}hkH^2 + 288\dot{A}kH^2h^2 + 576hkA^3H^2 + 288\dot{h}kA^2H^2 + 1008kA^2H^2h^2 \\ &+ 576AkH^2h^3 - 144\dot{A}H^4k^2 - 864\dot{h}A^2h^2 - 216A^2H^4k^2 - 576A\dot{h}h^3 - 576A\dot{A}h^3 \\ &- 144\dot{A}h^4 - 576A^3h^3 - 1368A^2h^4 - 576Ah^5\end{aligned}$$

$$\begin{aligned}E_{(3)}{}^{x_0}{}_{x_0} =& 432A\dot{A}hkH^2 + 648hkA^3H^2 + 216\dot{h}kA^2H^2 + 216kA^2H^2h^2 + 216\dot{A}kA^2H^2 + 216kA^4H^2 \\ &- 432h\dot{h}A^3 - 648\dot{h}A^2h^2 - 648\dot{A}A^2h^2 - 432A\dot{A}h^3 - 648A^4h^2 - 1224A^3h^3 \\ &- 648A^2h^4\end{aligned}$$

5.3 The background field equations

With the author's weights and $\Lambda = -10H^2$ the three independent components are (DERIVED; the x_4 component is a constraint, the other two are linear in $\dot{A}$ and $\dot{h}$)

$$\begin{aligned}x_4: \quad & -\frac{15A^2h^4}{16H^4} - \frac{5A^3h^3}{4H^4} - \frac{3h^4}{8H^2} - \frac{9Ah^3}{2H^2} - \frac{27A^2h^2}{4H^2} + \frac{9kA^2h^2}{8H^2} \\ & - \frac{3hA^3}{2H^2} + \frac{3hkA^3}{4H^2} - 6h^2 + \frac{3kh^2}{4} - 12Ah + \frac{9Ahk}{2} \\ & - 3A^2 + \frac{9kA^2}{4} - \frac{3A^2k^2}{16} - 10H^2 + 6kH^2 - \frac{3H^2k^2}{8} \\ & = -\kappa\rho\end{aligned}$$

$$\begin{aligned}x_1: \quad & \Big[-\frac{h^4}{8H^4} - \frac{Ah^3}{2H^4} - \frac{3h^2}{2H^2} + \frac{kh^2}{4H^2} - \frac{Ah}{H^2} + \frac{Ahk}{2H^2} \\ & \quad - 2 + \frac{3k}{2} - \frac{k^2}{8} \Big] \dot{A} \\ & + \Big[-\frac{Ah^3}{2H^4} - \frac{3A^2h^2}{4H^4} - \frac{3h^2}{2H^2} - \frac{3Ah}{H^2} + \frac{Ahk}{2H^2} - \frac{A^2}{2H^2} \\ & \quad + \frac{kA^2}{4H^2} - 4 + \frac{3k}{2} \Big] \dot{h} \\ & - \frac{Ah^5}{2H^4} - \frac{19A^2h^4}{16H^4} - \frac{A^3h^3}{2H^4} - \frac{15h^4}{8H^2} - \frac{6Ah^3}{H^2} + \frac{Akh^3}{2H^2} \\ & - \frac{17A^2h^2}{4H^2} + \frac{7kA^2h^2}{8H^2} - \frac{hA^3}{H^2} + \frac{hkA^3}{2H^2} - 10h^2 + \frac{9kh^2}{4} \\ & - 8Ah + 3Ahk - 3A^2 + \frac{9kA^2}{4} - \frac{3A^2k^2}{16} - 10H^2 \\ & + 6kH^2 - \frac{3H^2k^2}{8} \\ & = \kappa p_3\end{aligned}$$

$$
\begin{aligned}
x_0:\ &\Big[-\frac{3Ah^3}{8H^4}-\frac{9A^2h^2}{16H^4}-\frac{9h^2}{8H^2}-\frac{9Ah}{4H^2}+\frac{3Ahk}{8H^2}-\frac{3A^2}{8H^2}\\
&\quad+\frac{3kA^2}{16H^2}-3+\frac{9k}{8}\Big]\dot A\\
&+\Big[-\frac{9A^2h^2}{16H^4}-\frac{3hA^3}{8H^4}-\frac{3h^2}{8H^2}-\frac{9Ah}{4H^2}-\frac{9A^2}{8H^2}+\frac{3kA^2}{16H^2}\\
&\quad-3+\frac{3k}{8}\Big]\dot h\\
&-\frac{9A^2h^4}{16H^4}-\frac{17A^3h^3}{16H^4}-\frac{9A^4h^2}{16H^4}-\frac{3h^4}{8H^2}-\frac{27Ah^3}{8H^2}-\frac{45A^2h^2}{8H^2}\\
&+\frac{3kA^2h^2}{16H^2}-\frac{27hA^3}{8H^2}+\frac{9hkA^3}{16H^2}-\frac{3A^4}{8H^2}+\frac{3kA^4}{16H^2}-6h^2\\
&+\frac{3kh^2}{8}-9Ah+\frac{9Ahk}{8}-6A^2+\frac{9kA^2}{4}-10H^2\\
&+3kH^2\\
&=\kappa\bar p
\end{aligned}
$$

Here $\bar p=(p_0+3p_t)/4$. Because $E^{x_0}{}_{x_0}=E^{x_5}{}_{x_5}$ identically, the order-0 metric admits only sources with $p_0=p_t$ (and no off-diagonal components); for a source with $p_0\neq p_t$ the variation of the action with respect to the common block scale f gives the equation above with $\bar p$, and p_0-p_t drives a block-shape mode outside the order-0 metric (OPEN). The source is conserved,

$$\dot\rho=-3A(\rho+p_3)-h(\rho+p_0)-3h(\rho+p_t)=-3A(\rho+p_3)-4h(\rho+\bar p).$$

The total source is the stabiliser ($\kappa\rho_b=\kappa\rho_b^*(k/k_*)^4$, $p_3=-\rho_b$, $\bar p=\rho_b$) plus matter.

5.4 The static block and the stabiliser

For $h=0$, $\dot A=0$:

$$x_4:\ -3G(k)A^2+6H^2k(1-2w_2k)+\Lambda+\kappa\rho=0,\qquad x_1:\ \text{the same with }\ -\kappa p_3\ \text{for}\ \kappa\rho,$$

$$x_0:\ 6(12w_2k-1)A^2-\frac{12(w_2-18w_3k)}{H^2}A^4+3H^2k+\Lambda-\kappa\bar p=0,\qquad G(k)=1-24w_2k+72w_3k^2=\frac{k^2-12k+16}{16}.$$

Without sources and with $A=0$ the only static block is $k_0=1/(4w_2)=8$, $\Lambda_0=-3H^2/(4w_2)=-24H^2$, with $G(k_0)=-1$ (repulsive four-dimensional gravity). With the stabiliser and $A=0$:

$$\kappa\rho_b=10H^2-6H^2k+\tfrac{3}{8}H^2k^2=\kappa\bar p=3H^2k-10H^2\implies k_\pm=12\pm\tfrac{4}{3}\sqrt{51},$$

$$k_*=12+\tfrac{4}{3}\sqrt{51}=21.5219,\quad f_*=0.215556,\quad G(k_*)=\tfrac{20}{3}+\sqrt{51}=13.8081,\quad \kappa\rho_b^*=(26+4\sqrt{51})H^2.$$

5.5 The dynamical system, the radion and the response to matter

The x_1 and x_0 components are solved for $\dot A$ and $\dot h$; with $\dot k=-2kh$ and the constraint they form a closed system for (A,h,k). Linearised about $(0,0,k_*)$ (DERIVED):

$$\delta\dot A=J_{Ak}\delta k,\quad \delta\dot h=J_{hk}\delta k,\quad \delta\dot k=-2k_*\delta h,\qquad \omega_r^2=2k_*J_{hk}=\Big(\frac{31552}{2249}-\frac{2416\sqrt{51}}{6747}\Big)H^2=(3.38705H)^2,$$

eigenvalues $0,\pm i\omega_r$: the block is stable and oscillates (the radion), the zero mode being the direction δA (a small four-dimensional cosmological term). For the stabiliser family $\bar p_b=\nu\rho_b$, $\rho_b\propto k^{2(1+\nu)}$, the static block is stable for every $\nu\in\{1/4,1/2,3/4,1,5/4,3/2,2,3\}$ ($\omega_r=3.33H$ to $4.10H$). With matter sources to first order (A^2 counts as first order: the constraint gives $3G(k_*)A^2=\kappa\delta\rho$ because $\frac{d}{dk}[6H^2k(1-2w_2k)+\kappa\rho_b]$ vanishes at k_*):

$$\delta\dot A=J_{Ak}\,\delta k+s_{A3}\,\kappa\delta p_3+s_{Ap}\,\kappa\delta\bar p+q_A\,\kappa\delta\rho,$$

$$\delta\dot h=J_{hk}\,\delta k+s_{h3}\,\kappa\delta p_3+s_{hp}\,\kappa\delta\bar p+q_h\,\kappa\delta\rho,$$

$$\delta\ddot{k} + \omega_r^2\,\delta k = -2k_*\big(s_{h3}\,\kappa\delta p_3 + s_{hp}\,\kappa\delta\bar{p} + q_h\,\kappa\delta\rho\big),$$

$$J_{Ak} = \tfrac{6(238-19\sqrt{51})}{2249}H^2, \quad J_{hk} = \tfrac{4063-485\sqrt{51}}{2249}H^2, \quad s_{A3} = \tfrac{2283}{4498} - \tfrac{162\sqrt{51}}{2249}, \quad s_{Ap} = \tfrac{28\sqrt{51}-114}{2249},$$

$$q_A = \tfrac{135021}{265382} - \tfrac{10308\sqrt{51}}{132691}, \quad s_{h3} = \tfrac{21\sqrt{51}}{2249} - \tfrac{171}{4498}, \quad s_{hp} = \tfrac{239\sqrt{51}}{6747} - \tfrac{485}{2249}, \quad q_h = \tfrac{57}{4498} - \tfrac{7\sqrt{51}}{2249}$$

(numerically $0.272956H^2$, $0.266522H^2$, -0.006853, 0.038221, -0.045997, 0.028666, 0.037320, -0.009555). In the adiabatic response ($\delta\ddot{k}=0$), $\delta k = (-0.107556\,\kappa\delta p_3 - 0.140028\,\kappa\delta\bar{p} + 0.035852\,\kappa\delta\rho)/H^2$, and 3-space obeys exactly

$$\delta\dot{A} = -\frac{\kappa(\delta\rho + \delta p_3)}{2G(k_*)},$$

the four-dimensional Friedmann acceleration equation (DERIVED).

5.6 The four-dimensional effective gravity

On a product of a four-dimensional metric with the static block (constant curvature $K=-H^2k$) the Lovelock densities factorise, $\mathcal{L}_{(p)} = \sum_{r+s=p} \frac{p!}{r!\,s!}\mathcal{L}^{(4)}_{(r)}\mathcal{L}^{(int)}_{(s)}$ ($p=1,2,3$ the Lovelock order), with $\mathcal{L}^{(int)}_{(s)} = (1, 12K, 24K^2, 0)$ for $s=0,1,2,3$, $\mathcal{L}^{(4)}_{(3)}\equiv 0$ and $\mathcal{L}^{(4)}_{(2)}$ topological in four dimensions, so

$$\sum_p \alpha_p\mathcal{L}_{(p)} - 2\Lambda - 2\kappa\rho_b = G(k)\,R^{(4)} + (\text{topological}) - 2G(k)\Lambda_4, \qquad \Lambda_4 = \frac{6H^2k(1-2w_2k) + \Lambda + \kappa\rho_b}{G(k)},$$

and the four-dimensional field equations are those of general relativity,

$$G^{(4)}_{\mu\nu} + \Lambda_4\,g^{(4)}_{\mu\nu} = 8\pi G_N\,T^{(4)}_{\mu\nu}, \qquad 8\pi G_N = \frac{\kappa}{G(k_*)V_\perp},$$

$$V_\perp = \frac{2f^4v_t(1-e^{-3\eta})}{3H}, \qquad T^{(4)}_{\mu\nu} = \int dx_0\,dx_5\,dx_6\,dx_7\sqrt{|g_\perp|}\,T_{\mu\nu},$$

with $\Lambda_4 = 0$ at the static point (the observed dark energy is a tuned residual, $\Lambda_4/H^2 \sim 10^{-92}$, OPEN). The gauge couplings are $g_i^2 = \hat{g}_i^2/V_\perp$.

6 The field equations in the hidden coordinate: the masses

On the static block every four-dimensional mass is an eigenvalue of the x_0 operator of its field (the author's requirement: the wave functions depend on x_0 periodically with period $\pi/(6H)$, through $y(z)$). With $\xi = y/L \in [-1,0]$ on each half of a period, bosonic eigenvalues are $\Lambda = (m\,fL)^2$ and fermionic ones $e = \varepsilon\,fL$.

6.1 Gauge fields and the graviton

A four-dimensional component $A_\mu(x)u(y)$ (gauge $A_y=0$) has the reduced action $\int_{-L}^0 dy\,f^4e^{3Hy}\big[(\partial_y u)^2/f^2 - m^2u^2\big]$ per half (times $1/\hat{g}^2$), hence (DERIVED)

$$-\frac{1}{f^2}e^{-3Hy}\partial_y\big(e^{3Hy}\partial_y u\big) = m^2u \iff -e^{-3\eta\xi}\big(e^{3\eta\xi}u'\big)' = \Lambda u,$$

$$\text{fold: } u'(0)=0 \text{ (even) or } u(0)=0 \text{ (odd)}; \qquad \text{tip: } u'(-1) = \beta\,u(-1),$$

the tip condition being the natural boundary condition of the tip mass term: $\beta=0$ for the photon and the gluons, β_W for W^+ and W^- (one value for both: they are particle and antiparticle), $\beta_Z = \beta_W/\cos^2\theta_W$ for the Z (the ratio of the coefficients $\hat{g}^2$ and $\hat{g}^2+\hat{g}'^2$ of the tip mass terms; θ_W is the angle of the couplings, $\cos^2\theta_W = \hat{g}^2/(\hat{g}^2+\hat{g}'^2)$, fixed in UM-2 by the measured masses and the measured electroweak parameter ρ_0 as $\cos^2\theta_W = (m_W^2/m_Z^2)/\rho_0$). The transverse-traceless graviton obeys the same operator with $\beta=0$ in Einstein gravity. With $u = e^{-3\eta\xi/2}\tilde{u}$, $q^2 = \Lambda - 9\eta^2/4$:

$$\text{even: } \Big[q^2 + \frac{3\eta}{2}\Big(\beta + \frac{3\eta}{2}\Big)\Big]\sin q - \beta q\cos q = 0, \qquad \text{odd: } q\cos q + \Big(\beta + \frac{3\eta}{2}\Big)\sin q = 0.$$

For $\beta=0$: $\Lambda=0$ (the flat profile; graviton, photon and gluons are exactly massless), $9\eta^2/4 + n^2\pi^2$ (even), $q\cot q = -3\eta/2$ (odd).

6.2 Fermions

With $\Psi = e^{ikx_1} e^{-\frac{3}{2}Hy} e^{a_4/2} b_4^{-1} V_\beta\, \chi(y, x_4)$ the eight-dimensional Dirac equation (canonical spin connection included) is EXACTLY (DERIVED, all blocks, general a_4, b_4, $M(y)$, $v(y)$)

$$\gamma^M D_M \Psi - \big(M(y) - iv(y)\gamma^{x_4}\big)\Psi = e^{ikx_1} e^{-\frac{3}{2}Hy} e^{a_4/2} b_4^{-1}\, \gamma^{x_4} V_\beta \big[\partial_4 \chi + i h_j \chi\big],$$

$$h_j = j\Big[-\frac{i}{f}\sigma_1 \partial_y + M(y)\sigma_2 + e^{-a_4} k\, \sigma_3 \Big] + v(y),$$

so $i\partial_4 \chi = h_j \chi$. In every block (j, s_2, s_3): $\gamma^{x_0} = \sigma_3$, $B = j s_2$, $C = s_2 \sigma_2$, $\Gamma_4 = -j s_2 \sigma_3$, $P_+ = (1 - B\sigma_3)/2$. For a Standard-Model multiplet $M = \mu_\psi$, $v = 0$; with $c = f\mu_\psi L$ the zero modes are $\chi = (e^{c\xi}, 0)$ (doublet, $\Gamma_4 = +1$) and $\chi = (0, e^{-c\xi})$ (singlet, $\Gamma_4 = -1$), with $h_j(e^{c\xi}, 0) = j e^{-a_4} k\, (e^{c\xi}, 0)$ exactly (massless four-dimensional particles). At $k = 0$ in the block $j = +1$ the components of $\chi = (a, ib)$ obey $a' = c\, a - e\, b$, $b' = e\, a - c\, b$ ($e = \varepsilon f L$); the doublet ($c_L > 0$) and singlet ($c_R < 0$) solutions that satisfy the fold conditions reach the tip with

$$(a_L, b_L) = (C_L - c_L s_L,\ -s_L e), \qquad (a_R, b_R) = (s_R e,\ C_R + c_R s_R), \qquad s = \frac{\sinh w}{w}, \quad C = \cosh w, \quad w = \sqrt{c^2 - e^2},$$

and the tip mixing conditions of one generation give the exact mass condition

$$D(e) = e^2 s_L s_R - \kappa_f^2\, (C_L - c_L s_L)(C_R + c_R s_R) = 0,$$

$$e \simeq \kappa_f\, \zeta(c_L)\, \zeta(-c_R), \qquad \zeta(c)^2 = \frac{c\, e^{-c}}{\sinh c} \qquad (c_L > 0,\ c_R < 0).$$

For three generations, with doublet amplitudes $\mathcal{A}_\alpha$ and singlet amplitudes $\mathcal{B}_i$, the matrix conditions $\mathrm{diag}(b_{L\alpha})\mathcal{A} = \kappa\, \mathrm{diag}(b_{Ri})\mathcal{B}$, $\mathrm{diag}(a_{Ri})\mathcal{B} = -\kappa^\dagger \mathrm{diag}(a_{L\alpha})\mathcal{A}$ give

$$\Big[\mathrm{diag}(b_{L\alpha}) + \kappa\, \mathrm{diag}\Big(\frac{b_{Ri}}{a_{Ri}}\Big)\kappa^\dagger\, \mathrm{diag}(a_{L\alpha})\Big]\mathcal{A} = 0,$$

the masses being the roots of its determinant and $\mathcal{A}$ the left-handed flavour content. For a common doublet bulk mass and a unitary κ (DERIVED)

$$\det\Big[\ldots\Big] = \prod_{i=1}^{3} \frac{b_L a_{Ri} + a_L b_{Ri}}{a_{Ri}}, \qquad b_L a_R + a_L b_R = -D(e)\big|_{\kappa_f = 1}, \qquad \mathcal{A}^{(i)} = \kappa\, \hat{e}_i, \quad \mathcal{B}^{(i)} \propto \hat{e}_i :$$

the masses are the one-generation roots with $\kappa_f = 1$, the left-handed rotation of each sector is its tip matrix (so that $U = \kappa_e^\dagger \kappa_\nu$ and $V = \kappa_u^\dagger \kappa_d$), and the right-handed rotations are trivial (verified with mpmath at 50 digits to 3.0×10^{-42}, the two sets of roots refined in the same computation; the masses stored by the mass computation agree with them to 4.5×10^{-11}).

7 dirac16complex: the Kohn-Sham equations coupled to all fields

The interacting many-body state of Ψ on the block is described by Kohn-Sham orbitals of

$$h_j \chi = \varepsilon \chi, \qquad M_{eff}(y) = m + \frac{15}{16}\lambda S(y) + \frac{\langle \mathcal{F} \rangle}{\Lambda_\star^3}, \qquad v(y) = -\frac{\lambda}{16} n(y) + \frac{c_V}{\Lambda_\star^6} \sum_{i=1}^{3} \sum_{\psi} q_\psi n_{\psi i}(y),$$

where $\langle \mathcal{F} \rangle = c_G \langle G^\alpha_{MN} G^{\alpha MN} \rangle + \ldots$ is the gauge-field condensate (in the QCD vacuum the gluon condensate, a uniform mass shift), $n_{\psi i}$ the number densities of the Standard-Model fermions of the three generations ($\langle \bar{\psi}_i \gamma^{x_4} \psi_i \rangle = i\, n_{\psi i}$; I11). The densities are

$$n = \sum_o w\, g\, f_o\, n_o, \quad S = \sum_o w\, g\, f_o\, s_o, \qquad n_o = P \chi_o^\dagger \chi_o, \quad s_o = P \chi_o^\dagger j \sigma_2 \chi_o, \quad t_o = P \chi_o^\dagger j \sigma_3 \chi_o, \quad P = \frac{e^{-3Hy}}{\mathrm{Vol}_7},$$

with $w = 1/2$ for each fold parity, $g = 4 r_3(|\vec{n}|^2)$ the degeneracy of the momentum shell $\vec{k} = \Delta k\, \vec{n}$ ($\vec{n} \in \mathbb{Z}^3$, $r_3(|\vec{n}|^2)$ the number of representations of $|\vec{n}|^2$ as a sum of three squares), $\mathrm{Vol}_7 = (2\pi/\Delta k)^3 v_t f^4$ (at $a_4 = 0$; v_t

the coordinate volume of the extra times) the normalisation volume of a cell, of dimension length^6 (the proper seven-volume of a cell, with its hidden extent, is $2\,\mathrm{Vol}_7(1-e^{-3\eta})/(3H)$), and occupations f_o (aufbau, or Mermin $f_o = 1/(e^{(\varepsilon_o-\mu)/T}+1)$). The energy and the energy-momentum tensor are

$$E_{KS} = \sum_o g\, f_o\, \varepsilon_o - 2\,\mathrm{Vol}_7 \int_{-L}^{0} e^{3Hy} e_{int}\, dy, \qquad e_{int} = \frac{15}{32}\lambda S^2 - \frac{1}{32}\lambda n^2,$$

$$\rho = \sum_o w g f_o \varepsilon_o\, n_o - e_{int}, \qquad p_3 = \sum_o w g f_o\, \frac{e^{-a_4|k|}}{3} t_o + e_{int}, \qquad p_t = e_{int},$$

$$p_0 = \sum_o w g f_o \big[(\varepsilon_o - v) n_o - M_{eff} s_o - e^{-a_4|k|} t_o\big] + e_{int},$$

which satisfy $E_{KS} = 2\mathrm{Vol}_7 \int e^{3Hy}\rho\, dy$ and the y-conservation law $\big[e^{3Hy}p_0\big]_{-L}^{0} = 3H\int_{-L}^{0} e^{3Hy} p_t\, dy$ (DERIVED for every self-consistent state). Boundary conditions: fold $\chi_2(0)=0$ (even) or $\chi_1(0)=0$ (odd), tip $\chi_2(-L)=0$ (from $(1\mp\gamma^{x_0})\Psi=0$). The Kohn-Sham source enters the background equations through its block averages ρ, p_3 and $\bar p = (p_0+3p_t)/4$.

8 The linear perturbations (order ϵ): CMB, CNB, BAO

The x_0-independent ($n=0$) perturbations obey the four-dimensional Einstein equations of section 5.6 linearised about a flat Friedmann background; the $n\geq 1$ modes (Kaluza-Klein gravitons and gauge bosons, mass ≥ 99 TeV) and the radion (48 TeV) are absent at cosmological scales. In conformal time τ ($dx_4 = a\,d\tau$, $a = e^{a_4}$, $\mathcal{H} = a'/a$) and the Newtonian gauge $ds^2 = a^2\big[-(1+2\psi)d\tau^2 + (1-2\phi)d\vec{x}^{\,2}\big]$ (UM-2: $\psi=\epsilon\Phi$, $\phi=\epsilon\Psi_N$), for a Fourier mode k (STANDARD, Ma and Bertschinger):

$$k^2\phi + 3\mathcal{H}(\phi' + \mathcal{H}\psi) = -4\pi G_N a^2 \delta\rho, \qquad k^2(\phi' + \mathcal{H}\psi) = 4\pi G_N a^2(\bar\rho + \bar P)\theta,$$

$$\phi'' + \mathcal{H}(\psi' + 2\phi') + (2\mathcal{H}' + \mathcal{H}^2)\psi + \frac{k^2}{3}(\phi-\psi) = 4\pi G_N a^2 \delta P, \qquad k^2(\phi-\psi) = 12\pi G_N a^2(\bar\rho+\bar P)\sigma,$$

summed over the species of UM-2:

- **dark matter** (the odd quanta of Ψ_A and Ψ_B, cold): $\delta_c' = -\theta_c + 3\phi'$, $\theta_c' = -\mathcal{H}\theta_c + k^2\psi$;
- **baryons** (u, d confined in protons and neutrons; electrons): $\delta_b' = -\theta_b + 3\phi'$, $\theta_b' = -\mathcal{H}\theta_b + c_s^2 k^2\delta_b + \frac{4\bar\rho_\gamma}{3\bar\rho_b} a n_e \sigma_T(\theta_\gamma - \theta_b) + k^2\psi$;
- **photons**: $\delta_\gamma' = -\frac{4}{3}\theta_\gamma + 4\phi'$, $\theta_\gamma' = k^2(\frac{1}{4}\delta_\gamma - \sigma_\gamma) + k^2\psi + a n_e\sigma_T(\theta_b - \theta_\gamma)$, $F_{\gamma 2}' = 2\sigma_\gamma' = \frac{8}{15}\theta_\gamma - \frac{3}{5}kF_{\gamma 3} - \frac{9}{5}a n_e\sigma_T\sigma_\gamma + \frac{1}{10}a n_e\sigma_T(G_{\gamma 0}+G_{\gamma 2})$, $F_{\gamma l}' = \frac{k}{2l+1}[lF_{\gamma(l-1)} - (l+1)F_{\gamma(l+1)}] - a n_e\sigma_T F_{\gamma l}$ ($l\geq 3$), $G_{\gamma l}' = \frac{k}{2l+1}[lG_{\gamma(l-1)} - (l+1)G_{\gamma(l+1)}] + a n_e\sigma_T[-G_{\gamma l} + \frac{1}{2}(F_{\gamma 2}+G_{\gamma 0}+G_{\gamma 2})(\delta_{l0} + \frac{\delta_{l2}}{5})]$;
- **the three neutrino mass eigenstates** ν_i, $i=1,2,3$ (masses $m_i = 1.000$, 8.712, 50.140 meV; after decoupling the flavour coherence is lost and each mass state propagates on its own; comoving momentum q, comoving energy $E_i = \sqrt{q^2 + a^2 m_i^2}$, Fermi-Dirac distribution $\bar f_\nu(q)$, multipoles $\mathcal{N}_l^{(i)}$ of the perturbed distribution): $\mathcal{N}_0^{(i)\prime} = -\frac{qk}{E_i}\mathcal{N}_1^{(i)} - \phi'\frac{d\ln\bar f_\nu}{d\ln q}$, $\mathcal{N}_1^{(i)\prime} = \frac{qk}{3E_i}(\mathcal{N}_0^{(i)} - 2\mathcal{N}_2^{(i)}) - \frac{E_i k}{3q}\psi\frac{d\ln\bar f_\nu}{d\ln q}$, $\mathcal{N}_l^{(i)\prime} = \frac{qk}{(2l+1)E_i}[l\mathcal{N}_{l-1}^{(i)} - (l+1)\mathcal{N}_{l+1}^{(i)}]$ ($l\geq 2$);
- the fold bands of Ψ are never thermalised and do not contribute: there is no dark radiation.

The background is $\mathcal{H}^2 = \frac{8\pi G_N}{3}a^2\bar\rho_{tot} + \frac{\Lambda_4}{3}a^2$, adiabatic initial conditions with the power spectrum $\mathcal{P}(k) = A_s(k/k_0)^{n_s-1}$. The relativistic content is $N_{eff} = 3.0432$, carried by the three mass eigenstates (each with the degeneracy $N_{eff}/3 = 1.0144$ at the temperature $T_\nu = (4/11)^{1/3}T_\gamma$ of instantaneous decoupling, which represents the spectrum heated by e^+e^- annihilation; the adopted $N_{eff} = 3.0432$ is the precision value, close to the 3.0476 to 3.0485 computed in section 9.4).

9 The flavour evolution of the three neutrinos

9.1 The four-fermion interactions at low energy

At momentum transfers far below m_W and m_Z the exchange of $W^\pm$ and Z between the zero-mode fermions (sections 2 and 4) reduces to current-current interactions. Written for the four-dimensional zero-mode fields in

four-component notation, with $P_L \equiv P_+ = (1+\Gamma_4)/2$ the projector on the chirality of the doublet zero modes and $P_R = (1-\Gamma_4)/2$, they are (STANDARD)

$$\mathcal{L}_{CC} = -2\sqrt{2}\,G_F\,(\bar\nu_\alpha\gamma^\mu P_L\ell_\alpha)(\bar\ell_\beta\gamma_\mu P_L\nu_\beta), \qquad \mathcal{L}_{NC} = -2\sqrt{2}\,G_F\,(\bar\nu_\alpha\gamma^\mu P_L\nu_\alpha)\sum_f \bar f\gamma_\mu(g_L^f P_L + g_R^f P_R)f,$$

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8m_W^2}, \qquad g_L^f = T_3^f - Q^f\sin^2\theta_W, \qquad g_R^f = -Q^f\sin^2\theta_W,$$

summed over the flavours $\alpha,\beta \in \{e,\mu,\tau\}$ and the fermions f of the medium (the neutrino background is the term $V_{\nu\nu}$ of section 9.4), with $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ the flavour fields (UM-2 uses the measured $G_F = 1.1663788\times10^{-5}$ GeV^{-2}). A neutrino crossing a medium scatters forward coherently on it; averaged over an unpolarised medium at rest, the interactions add to its Hamiltonian the flavour-diagonal potentials

$$V_\alpha = \sqrt{2}\,G_F\Big[\delta_{\alpha e}\,(n_{e^-} - n_{e^+}) + \sum_f g_V^f\,(n_f - n_{\bar f})\Big], \qquad g_V^f = g_L^f + g_R^f = T_3^f - 2Q^f\sin^2\theta_W,$$

and in electrically neutral matter of electrons, protons and neutrons ($g_V^p = -g_V^e$, $g_V^n = -\frac12$)

$$V = \mathrm{diag}(V_{CC},0,0) + V_{NC}\,1, \qquad V_{CC} = \sqrt{2}\,G_F n_e, \qquad V_{NC} = -\frac{G_F n_n}{\sqrt{2}}.$$

Antineutrinos have $-V$.

9.2 Vacuum and matter

The flavour amplitudes ψ_α of a neutrino of energy $E \gg m_i$ evolve along its path as

$$i\frac{d\psi}{dx_4} = \mathcal{H}\psi, \qquad \mathcal{H} = \frac{1}{2E}\,U\,\mathrm{diag}(m_1^2,m_2^2,m_3^2)\,U^\dagger + V,$$

with $U \to U^*$ and $V \to -V$ for antineutrinos (V_{NC} is a multiple of the identity and drops out). For a constant potential the evolution operator is $S(L) = e^{-i\mathcal{H}L}$ and $P(\nu_\alpha\to\nu_\beta) = |S_{\beta\alpha}|^2$; for a varying one it is the ordered product of the operators of thin layers. In vacuum

$$P(\nu_\alpha\to\nu_\beta) = \Big|\sum_i U_{\beta i}U_{\alpha i}^*\,e^{-im_i^2L/(2E)}\Big|^2,$$

$$P(\nu_\mu\to\nu_e) - P(\nu_e\to\nu_\mu) = -16J\sin\Delta_{21}\sin\Delta_{31}\sin\Delta_{32}, \qquad \Delta_{ij} = \frac{\Delta m_{ij}^2 L}{4E},$$

with the Jarlskog invariant $J = \mathrm{Im}(U_{e2}U_{\mu3}U_{e3}^*U_{\mu2}^*) = c_{12}s_{12}c_{23}s_{23}c_{13}^2s_{13}\sin\delta$ and CPT, $P(\nu_\alpha\to\nu_\beta) = \bar P(\bar\nu_\beta\to\bar\nu_\alpha)$. In matter the mixing is resonant (the MSW effect) where the potential cancels the vacuum splitting of a sector: for the 1-3 sector, after the rotation R_{23} (which commutes with V),

$$2EV_{CC} = \Delta m_{ee}^2\cos2\theta_{13}, \qquad \Delta m_{ee}^2 = c_{12}^2\Delta m_{31}^2 + s_{12}^2\Delta m_{32}^2,$$

and for the 1-2 sector, after the rotation by θ_{13}, which leaves the potential $V_{CC}c_{13}^2$,

$$2EV_{CC}c_{13}^2 = \Delta m_{21}^2\cos2\theta_{12}.$$

A neutrino produced at a density where the effective 1-2 mixing angle is θ_{12}^m and propagating adiabatically to the vacuum arrives, averaged over the incoherent mass states, with

$$P_{ee} = c_{13}^4\Big(\frac12 + \frac12\cos2\theta_{12}\cos2\theta_{12}^m\Big) + s_{13}^4, \qquad \cos2\theta_{12}^m = \frac{\cos2\theta_{12} - A}{\sqrt{(\cos2\theta_{12}-A)^2 + \sin^2 2\theta_{12}}}, \qquad A = \frac{2EV_{CC}c_{13}^2}{\Delta m_{21}^2}.$$

A resonance at the density $n_e = \Delta m^2\cos2\theta/(2\sqrt{2}\,G_F E)$ is crossed adiabatically if the adiabaticity

$$\gamma = \frac{\Delta m^2\sin^2 2\theta}{2E\cos2\theta\,|d\ln n_e/dr|} \gg 1;$$

the probability of a jump between the matter eigenstates is the Landau-Zener expression $P_{LZ} = e^{-\pi\gamma/2}$.

9.3 The thermal potential

In the early Universe the electrons and positrons are relativistic and the potential is dominated by the thermal term of the W exchange (STANDARD; to leading order in T/m_W),

$$V_e - V_\mu = -\frac{8\sqrt{2}\,G_F p}{3m_W^2}\,\rho_{e^\pm}, \qquad \rho_{e^\pm} = \frac{7\pi^2}{60}T^4,$$

for a neutrino of momentum p; it suppresses the mixing as long as it exceeds $\Delta m^2\cos 2\theta/(2p)$.

9.4 The quantum kinetic equations and neutrino decoupling

In a gas of neutrinos the state of the neutrinos of momentum $\vec{p}$ is a 3×3 density matrix $\rho_{\vec{p}}$ (diagonal elements the occupation numbers of the flavours, off-diagonal ones the flavour coherence), and $\bar{\rho}_{\vec{p}}$ for antineutrinos. They obey (STANDARD)

$$i(\partial_{x_4} + \vec{v}_p\cdot\nabla)\rho_{\vec{p}} = \big[\Omega_{\vec{p}} + V + V_{\nu\nu}, \rho_{\vec{p}}\big] + i\,\mathcal{C}[\rho], \qquad \Omega_{\vec{p}} = \frac{U\,\mathrm{diag}(m_i^2)\,U^\dagger}{2p},$$

$$V_{\nu\nu} = \sqrt{2}\,G_F\int\frac{d^3q}{(2\pi)^3}\,(1-\vec{v}_p\cdot\vec{v}_q)\big(\rho_{\vec{q}} - \bar{\rho}_{\vec{q}}\big), \qquad \vec{v}_p = \frac{\vec{p}}{p},$$

with $\mathcal{C}$ the collision term of $\mathcal{L}_{CC}$ and $\mathcal{L}_{NC}$ (in an expanding Universe $\partial_{x_4}\to\partial_{x_4} - Hp\,\partial_p$). During neutrino decoupling the energy exchanged between the plasma and the neutrinos is followed with one temperature per species (STANDARD; Escudero 2020): with the photon temperature T_γ, the neutrino temperatures T_{ν_e} and $T_{\nu_\mu} = T_{\nu_\tau}$, the Hubble rate $H = \sqrt{8\pi\rho/3}/M_{Pl}$ of the total density (here H is not the author's constant; $M_{Pl} = G_N^{-1/2} = 1.22091\times 10^{19}$ GeV), the energy densities $\rho_\gamma = \pi^2T_\gamma^4/15$ and $\rho_{\nu_\alpha} = 7\pi^2T_{\nu_\alpha}^4/120$ per flavour and the $e^\pm$ energy density ρ_e and pressure p_e (with the electron mass, $g=4$, no chemical potential),

$$\frac{dT_\gamma}{dx_4} = -\frac{4H\rho_\gamma + 3H(\rho_e+p_e) + 3HT_\gamma\frac{dP_{int}}{dT_\gamma} + \frac{\delta\rho_{\nu_e}}{\delta x_4} + 2\frac{\delta\rho_{\nu_\mu}}{\delta x_4}}{\frac{d\rho_\gamma}{dT_\gamma} + \frac{d\rho_e}{dT_\gamma} + T_\gamma\frac{d^2P_{int}}{dT_\gamma^2}}, \qquad \frac{dT_{\nu_\alpha}}{dx_4} = -HT_{\nu_\alpha} + \frac{\delta\rho_{\nu_\alpha}/\delta x_4}{d\rho_{\nu_\alpha}/dT_{\nu_\alpha}},$$

$$\frac{\delta\rho_{\nu_e}}{\delta x_4} = \frac{G_F^2}{\pi^5}\Big[4(g_{eL}^2+g_{eR}^2)F(T_\gamma,T_{\nu_e}) + 2F(T_{\nu_\mu},T_{\nu_e})\Big], \qquad \frac{\delta\rho_{\nu_\mu}}{\delta x_4} = \frac{G_F^2}{\pi^5}\Big[4(g_{\mu L}^2+g_{\mu R}^2)F(T_\gamma,T_{\nu_\mu}) - F(T_{\nu_\mu},T_{\nu_e})\Big],$$

$$F(T_1,T_2) = 32f_a(T_1^9 - T_2^9) + 56f_sT_1^4T_2^4(T_1-T_2), \qquad f_a = 0.884, \quad f_s = 0.829$$

(f_a, f_s the Fermi-Dirac factors of the annihilation and scattering rates; $g_{eL} = \frac12 + s^2$, $g_{eR} = s^2$ for ν_e, which interacts with electrons through both currents, and $g_{\mu L} = -\frac12 + s^2$, $g_{\mu R} = s^2$ for $\nu_{\mu,\tau}$, with $s^2 = \sin^2\theta_W$, or the radiatively corrected values 0.727, 0.233, −0.273, 0.233). When oscillations equalise the flavours, one temperature T_ν obeys $dT_\nu/dx_4 = -HT_\nu + (\delta\rho_{\nu_e}/\delta x_4 + 2\delta\rho_{\nu_\mu}/\delta x_4)/(3\,d\rho_\nu/dT_\nu)$. The finite-temperature QED corrections enter through the interaction pressure of the $e^\pm\gamma$ plasma, $P_{int} = P_2 + P_3$, and its energy density $\rho_{int} = -P_{int} + T\,dP_{int}/dT$ (STANDARD; Bennett et al. 2020),

$$P_2 = -\frac{e^2T^2}{12\pi^2}J_1 - \frac{e^2}{8\pi^4}J_1^2 + \frac{e^2m_e^2}{16\pi^4}J_2, \qquad P_3 = \frac{e^3T}{12\pi^4}I^{3/2},$$

$$J_1 = \int_0^\infty dp\,\frac{p^2}{E}n_D, \qquad J_2 = \int_0^\infty\int_0^\infty dp\,dq\,\frac{pq}{E_pE_q}\ln\Big|\frac{p+q}{p-q}\Big|n_D(p)n_D(q),$$

$$I = \int_0^\infty dp\,\frac{p^2+E^2}{E}n_D, \qquad n_D = \frac{2}{e^{E/T}+1}, \qquad E = \sqrt{p^2+m_e^2},$$

whose massless limits are $P_2 = -5e^2T^4/288$ and $P_3 = e^3T^4/(36\sqrt{3}\,\pi)$. The effective number of neutrino species is

$$N_{eff} = \Big(\frac{11}{4}\Big)^{4/3}\Big[\Big(\frac{T_{\nu_e}}{T_\gamma}\Big)^4 + 2\Big(\frac{T_{\nu_\mu}}{T_\gamma}\Big)^4\Big]$$

after e^+e^- annihilation. For instantaneous decoupling at T_d the plasma conserves its entropy a^3s, $s = (\rho_\gamma + \frac13\rho_\gamma + \rho_e + p_e + \rho_{int} + P_{int})/T$, while $T_\nu\propto 1/a$.

9.5 The dense neutrino gas: the fast flavour instability

Near a supernova core $V_{\nu\nu}$ dominates $\Omega_{\vec{p}}$. For a homogeneous, axially symmetric gas of ν_e and $\bar{\nu}_e$ with the angular distributions $g_{\nu_e}(u)$, $g_{\bar{\nu}_e}(u)$ ($u = \cos\vartheta$ along the symmetry axis, $\int_{-1}^{1} g\, du/2 = 1$), the electron-lepton-number (ELN) distribution

$$G(u) = \sqrt{2}\, G_F \left[n_{\nu_e} g_{\nu_e}(u) - n_{\bar{\nu}_e} g_{\bar{\nu}_e}(u) \right]$$

governs the linearised evolution of the small flavour coherence S_u (the ex element of ρ, x a heavy-lepton flavour). With $v = (1, u)$, $\Lambda^\mu = \int du/2\, G(u)\, v^\mu$ and modes $S_u \propto e^{-i(\Omega x_4 - K x_3)}$ (x_3 the symmetry axis) in the shifted variables $\omega = \Omega - \Lambda^0$, $k = K - \Lambda^3$, the linear equations have non-trivial solutions if (STANDARD)

$$\det \Pi(\omega, k) = 0, \qquad \Pi^{\mu\nu} = \eta^{\mu\nu} + \int_{-1}^{1} \frac{du}{2}\, G(u)\, \frac{v^\mu v^\nu}{\omega - ku} \qquad (\eta = \mathrm{diag}(1, -1) \text{ on } (x_4, x_3)),$$

which splits into the axisymmetric branch and the azimuthal branch

$$(1 + J_0)(-1 + J_2) - J_1^2 = 0, \quad J_n = \int_{-1}^{1} \frac{du}{2}\, \frac{G(u)\, u^n}{\omega - ku}; \qquad -1 + \int_{-1}^{1} \frac{du}{2}\, \frac{G(u)\,(1 - u^2)}{2(\omega - ku)} = 0.$$

A root with $\mathrm{Im}\,\omega > 0$ for real k is an instability with the growth rate $\mathrm{Im}\,\omega$. The homogeneous mode ($K = 0$, i.e. $k = -\Lambda^3$) obeys

$$i \frac{dS_u}{dx_4} = (\Lambda^0 - u\Lambda^3)\, S_u - \int_{-1}^{1} \frac{du'}{2}\, G(u')\,(1 - uu')\, S_{u'},$$

and for the mode $K = \Lambda^3$ ($k = 0$) the dispersion relation is the quadratic $\omega^2 - (I_2 - I_0)\omega - (I_0 I_2 - I_1^2) = 0$, $I_n = \int du/2\, G\, u^n$, unstable with the growth rate $\frac{1}{2}\sqrt{4I_1^2 - (I_0 + I_2)^2}$ when this is real. An instability exists only if $G(u)$ changes sign (an ELN crossing; Morinaga 2022).

9.6 The cosmic neutrino background

After decoupling the neutrinos keep the Fermi-Dirac distribution $f(p) = 1/(e^{p/T_\nu} + 1)$ of their momentum, redshifted with $T_\nu \propto 1/a$, and lose their flavour coherence: the background is an incoherent mixture of the mass states, each with the number density and the energy density

$$n_i = 2 \int \frac{d^3p}{(2\pi)^3} f(p) = \frac{3\zeta(3)}{2\pi^2} T_\nu^3, \qquad \rho_i = 2 \int \frac{d^3p}{(2\pi)^3} \sqrt{p^2 + m_i^2}\, f(p)$$

(the factor 2 counts ν_i and $\bar{\nu}_i$, one helicity each), and the mean velocity $\langle v \rangle_i = \int p^2\, (p/\sqrt{p^2 + m_i^2}) f\, dp / \int p^2 f\, dp$. The flavour content of the mass state ν_i is $|U_{\alpha i}|^2$.

10 The coupled system and how it is solved

The fields couple as follows: gravity to everything (the canonical spin connection for Ψ and the fermions, the metric in every kinetic term, the stabiliser); Ψ to gravity, to itself (λ), to the fermion currents (I11) and to the gauge-field invariants (I12); the fermions of the three generations to the gauge fields, to Σ at the tip (through the tip matrices, which carry the flavour mixing) and to Ψ (I11); the gauge fields to the fermions, to Σ at the tip and to Ψ through I12 (a field-dependent coupling). UM-2 solves the coupled system in the following hierarchy, each step feeding the next:

1. the background: the static block $(A, h, k) = (0, 0, k_*)$ with the stabiliser, its stability and its first-order response to every block-averaged source; the primordial approach to it from the full nonlinear equations;
2. the x_0 eigenvalue problems on the static block: all masses and the mixing matrices (gauge bosons, graviton, the twelve fermions with their tip matrices, Ψ);
3. the Kohn-Sham equations of Ψ with the portal mean fields of the other fields and the self-interaction; the Kohn-Sham energy-momentum tensor returned to step 1 (the adiabatic block response);
4. the flavour evolution of the neutrinos (section 9) in vacuum, in matter, in the early Universe (decoupling and N_{eff}, which feed step 5) and in dense neutrino gases;
5. the four-dimensional effective theory and its linear perturbations (CMB, CNB, BAO).

11 What was verified

- the Lovelock tensors of the order-0 metric by four routes, the Bianchi identities ($k = 1, 2, 3$), the record's equations and the minisuperspace variation (sympy, exact);
- the static block, the stabiliser, the radion, the stabiliser family, the first-order response and the 4D Friedmann acceleration equation, the dimensional reduction (sympy, exact);
- the exact Dirac reduction in all eight blocks, the boundary conditions, the zero modes, the tip mixing condition of one generation and of three generations for an arbitrary complex tip matrix, the gauge x_0 operator and the gauge invariance of the tip terms with arbitrary generation matrices;
- the x_0 spectra (mpmath; finite elements to 5×10^{-10}; shooting to 6×10^{-10}); the three-generation tip determinant (mpmath, 50 digits: its roots equal the one-generation roots to 3.0×10^{-42}, its null vectors the columns of the tip matrix); the absence of flavour-changing Kaluza-Klein couplings (10^{-17});
- the gauge anomalies of the three generations (exact fractions);
- the flavour evolution (exact evolution operators against the vacuum formula to 3.9×10^{-15}, unitarity, CPT, the T-odd difference; the numerical solar evolution against the adiabatic formula to 1.3×10^{-3}; the exact 1-3 resonance against its formula to 1.6×10^{-4}; the massless limits of the QED integrals; the decoupling equations against the published values of the same method; three independent evaluations of the fast-flavour growth rates to 1.2×10^{-14});
- the Kohn-Sham solver (record reference states to 10^{-13}; energy identity; conservation law; Richardson; Hellmann- Feynman for every portal mean field).